Question Paper

Exam Date & Time: 20-Nov-2024 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. (ELECTRONICS & COMMUNICATION ENGINEERING) EXAMINATIONS - NOVEMBER / DECEMBER 2024
SUBJECT: ECE 2122/ECE_2122- NETWORK ANALYSIS

Marks: 50 **Duration: 180 mins.**

Answer all the questions.

1A) Determine the current in the 3 Ω resistor of the network shown in Fig. 1A. (4)

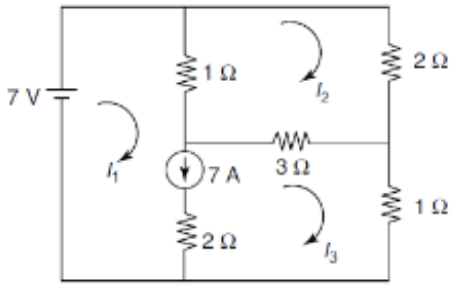


Fig: 1A

1B) Obtain the Thevenin's equivalent network of Fig. 1B. (4)

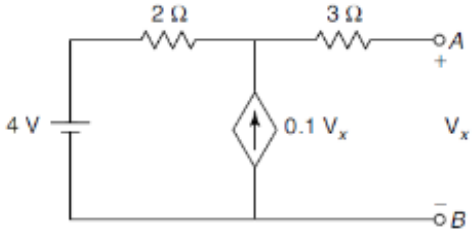


Fig: 1B

1C) Calculate current I and verify the reciprocity theorem for the network shown in Fig. 1C. (2)

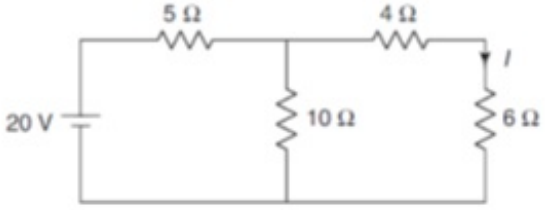


Fig: 1C

2A) In the network shown in Fig. 2A, the switch is closed at t = 0. With the capacitor uncharged, obtain the value for the following at $t = 0^+$ (5)

a) i
b) di/dt
c) d^2i/dt^2

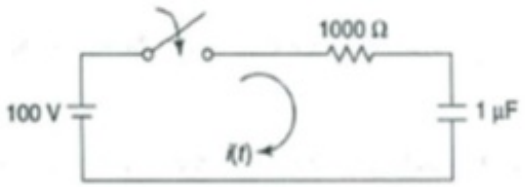


Fig: 2A

2B) For the differential equation given below, obtain the particular solution for v(t). (3)
$d^2v/dt^2 + 5dv/dt + 6v = e^{-2t}+5e^{-3t}$

2C) For the network shown in Fig. 2C, the switch is opened at t = 0. Calculate v(0+) for t > 0. (2)

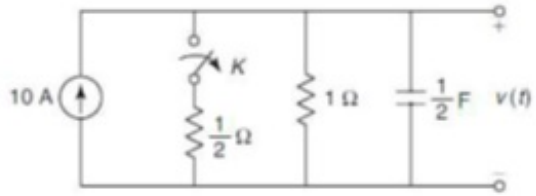
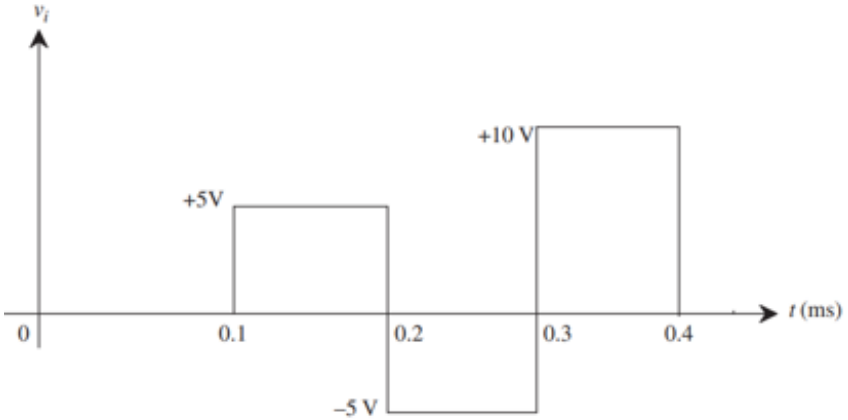
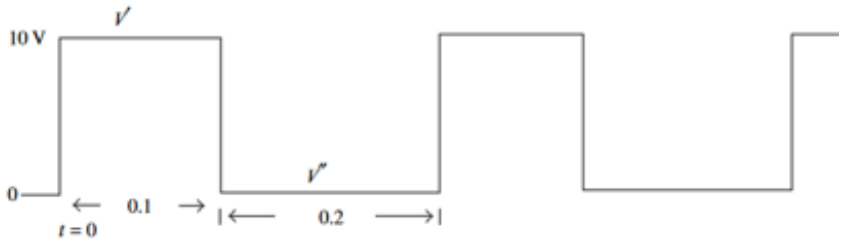


Fig: 2C

3A) The input signal shown in the figure is applied to an RC high pass circuit whose time constant is 0.4 ms. Draw the output waveform and mark all voltages, assuming that the capacitor is initially uncharged. (4)

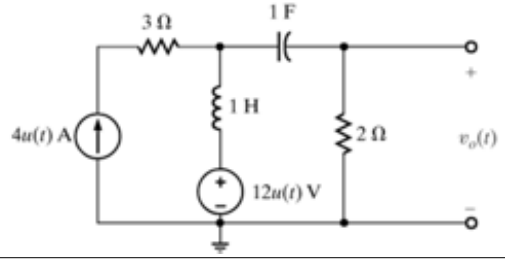


3B) The periodic waveform applied to an RC low-pass circuit is a square wave with T1 = 0.1 s, T2 = 0.2 s and time constant = 0.1 s. Calculate the output voltages and draw the output waveform. (4)

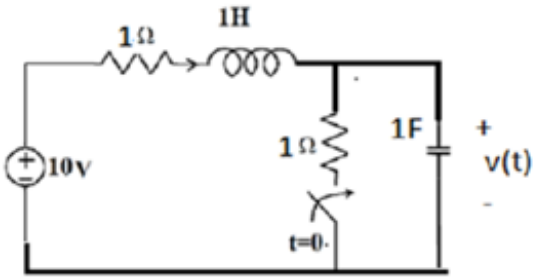


3C) A 20-kHz square wave is applied to an RC differentiating circuit. With R = 4 kΩ, it produces the output with a tilt of 5 per cent. Calculate the value of the capacitor and lower 3-dB frequency. (2)

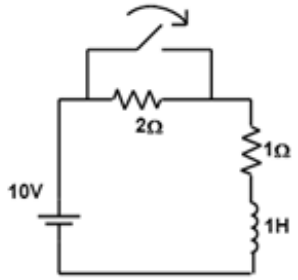
4A) Determine the voltage $v_o(t)$ in the circuit shown in figure using Laplace transform. (4)



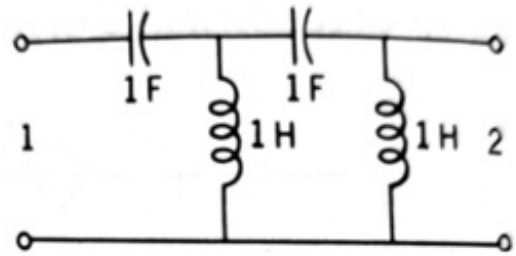
4B) Determine the voltage ***v(t)*** in the circuit shown in figure using Laplace transform. (4)



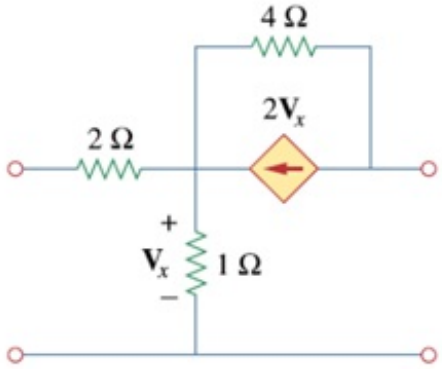
4C) In the circuit shown in figure, the switch is closed at t = 0., before which steady state has been reached. Find the expression for current in the circuit for t ≥ 0 using Laplace transform method. (2)



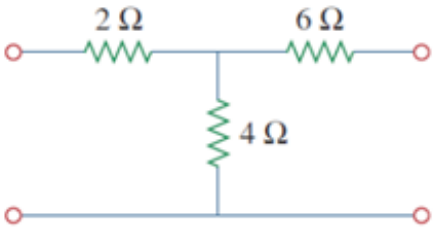
5A) For the network given in figure, terminate port 2 in a 1-Ω resistor and connect a voltage source at port 1. Let I_1 be the current in the voltage source and I_2 be the current in the 1-Ω load. Assign reference direction for each. For this network, compute $G_{12}= V_2/V_1$ and $\alpha_{12}= I_2/I_1$. (3)



5B) For the two-port network shown in figure, determine Y parameters (4)



5C) For the two-port network shown in figure, determine Transmission parameters. (3)



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